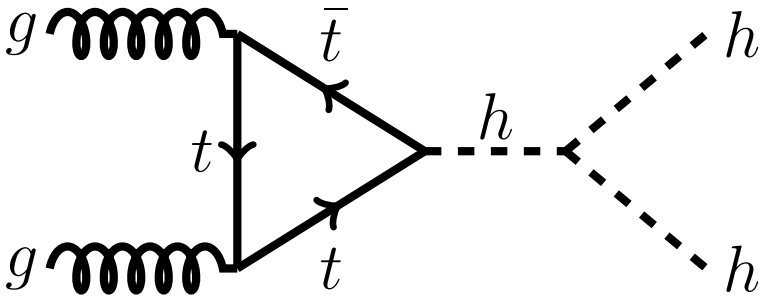
Probing the Higgs Potential with di-Higgs Production

Measuring the Tri-Linear Coupling at Particle Colliders



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Stony Brook University

GRADTALKS – May 12, 2017

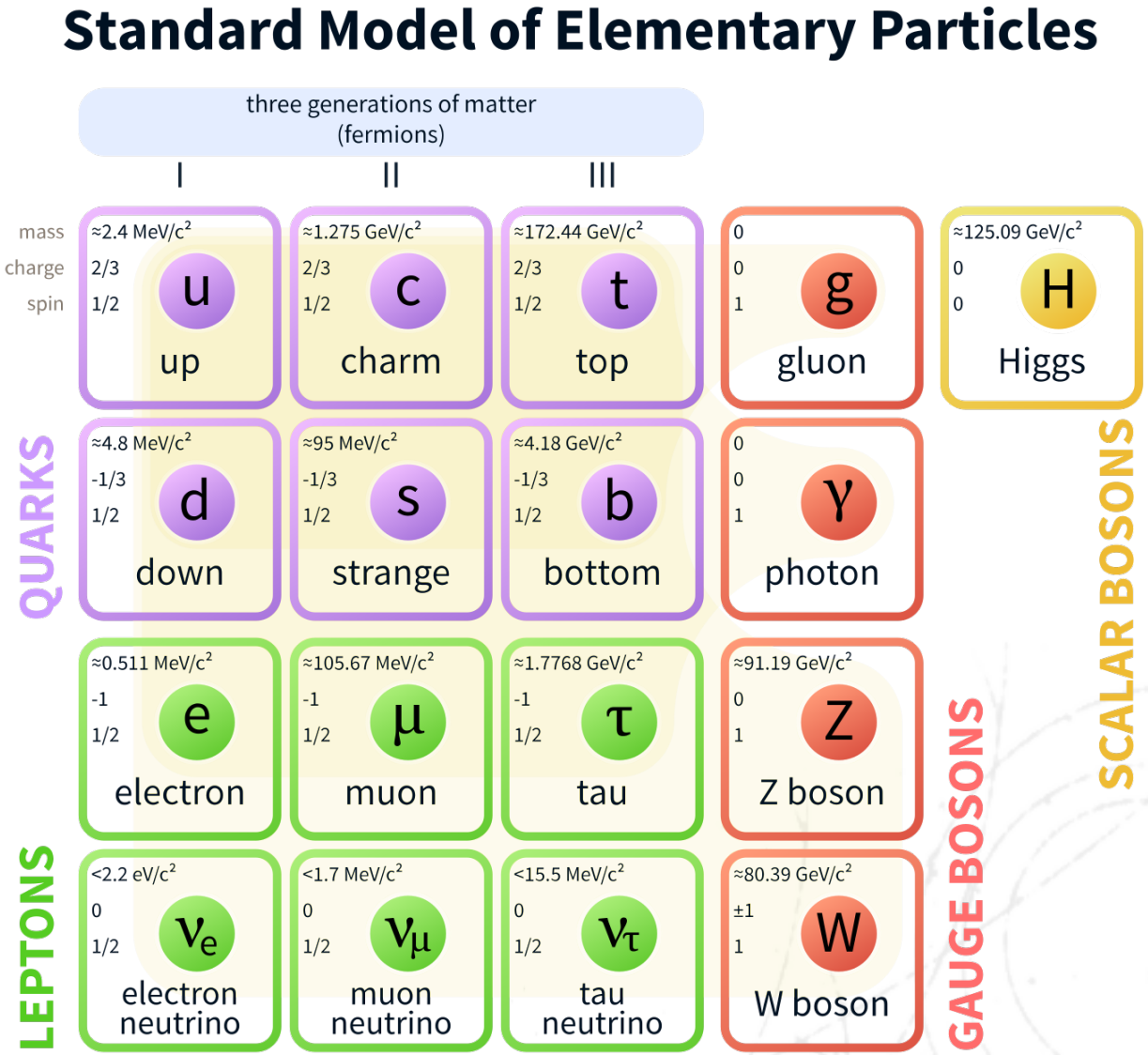
Outline

- **The Standard Model Higgs.**
  The Standard Model, the Higgs potential,
  spontaneous symmetry breaking, and Higgs interactions.

- **The Electroweak Phase Transition.**
  The finite temperature Higgs potential, behavior of the
  phase transition, and implications for baryogenesis.

- **Measuring di-Higgs production at colliders.**
  Higgs decays and detection, LHC phenomenology,
  and prospects at colliders

The Standard Model

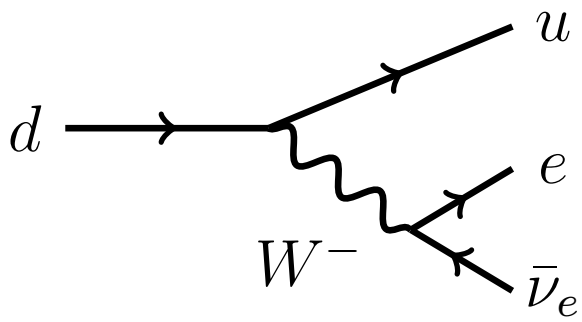


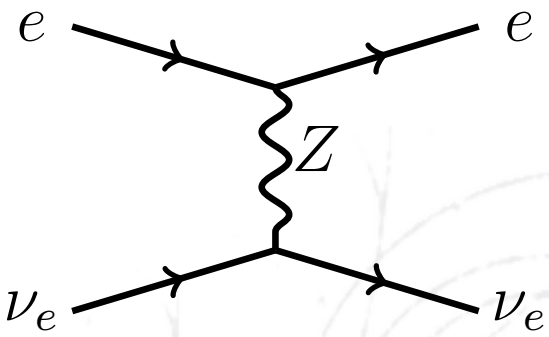
The Standard Electroweak Theory

The Standard Model is a Gauge Theory

$$G_{SM} = SU(3)_c \times SU(2)_L \times U(1)_Y$$

Describes electroweak interactions (e.g., β decay!)





But $W^{\pm}$, Z boson masses are forbidden by gauge symmetry...

$\Longrightarrow$ spontaneous symmetry breaking!

The Higgs Potential

Mexican Hats!

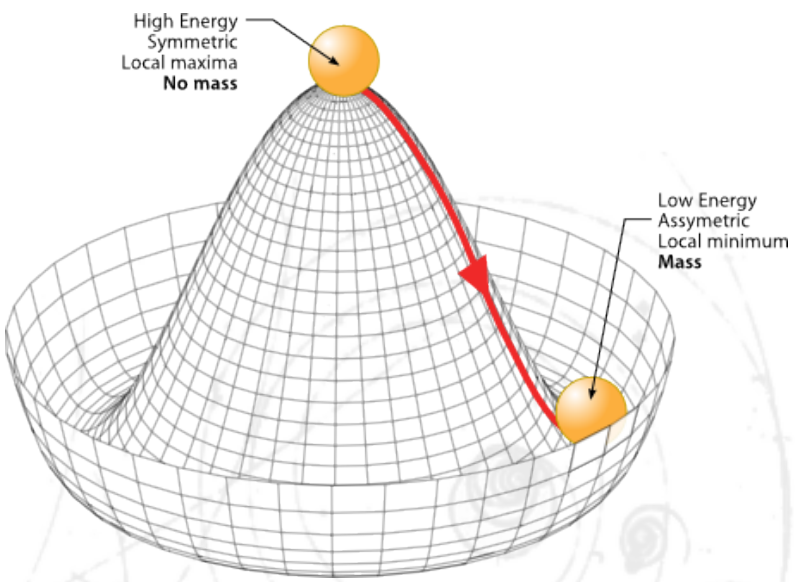
In the Standard Model, the Higgs potential takes a simple form:

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2$$

where

$$\phi(x) = \begin{pmatrix} w^+ \\ \frac{1}{\sqrt{2}}(v + h + iw^0) \end{pmatrix}$$

and $v = \sqrt{\mu^2/2\lambda}$.



quantum-bits.org

Higgs Potential in "Unitary Gauge"

In unitary gauge, $w^{\pm}, w^0$ become the longitudinal polarizations of $W^{\pm}, Z$.

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

The Higgs potential becomes:

$$V(h) = -\frac{1}{2} m_h^2 h^2 - \frac{\lambda}{2} m_h h^3 - \frac{1}{4} \lambda h^4$$

where $m_h = \sqrt{2}\mu = \sqrt{2\lambda} v$.

Note:

There are only two free parameters: $m_h = 125$ GeV and $v = 246 \sim$ GeV, both have been measured!

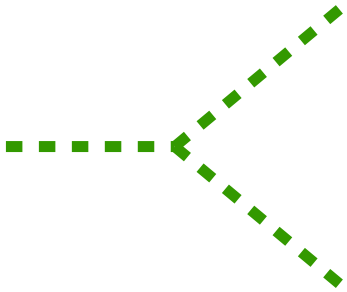
Higgs Interactions

What's the Big Deal with Scalars?

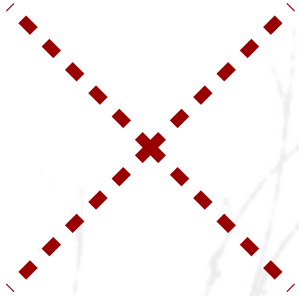
The Higgs is the only particle with self-interactions:

$$V_{\text{int}}(h) = \frac{\lambda_3}{3!}h^3 + \frac{\lambda_4}{4!}h^4$$

where in the SM, $\lambda_3 = 3m_h^2/v, \quad \lambda_4 = 3m_h^2/v^2$.

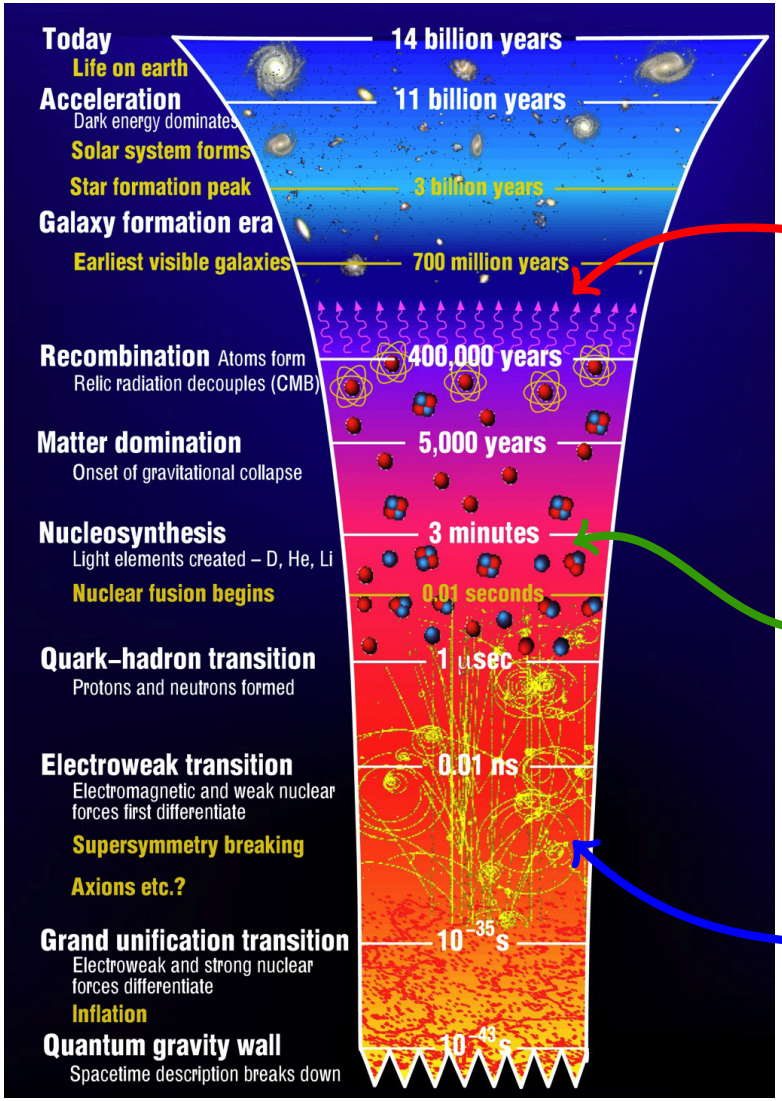


λ_3



λ_4

The Higgs and Cosmology



ctc.cam.ac.uk

CMB Measurements (Planck, etc.)

$T \sim 1$ eV

Light Element Abundances (He fraction, etc.)

$T \sim 1$ MeV

EWPT (Particle Colliders!)

$T \sim 100$ GeV

The Electroweak Phase Transition

In the early universe interactions with the plasma give the Higgs an effective potential:

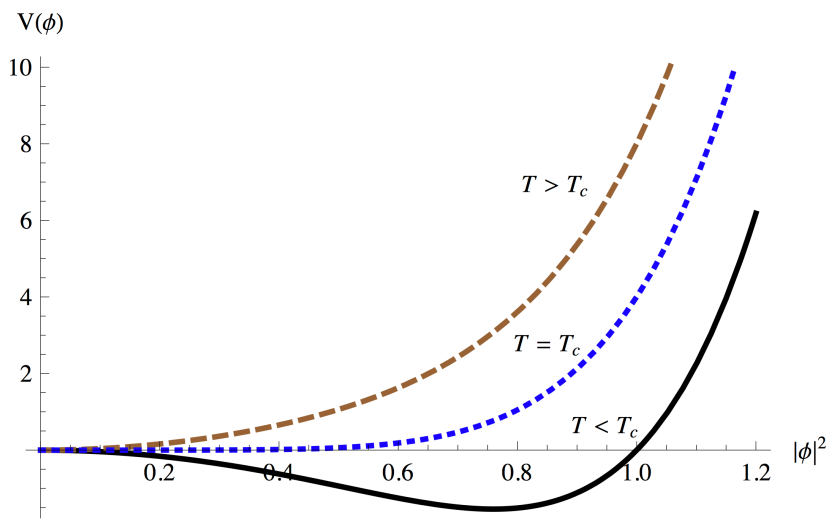
$$V_{\text{eff}}(h,T) \approxeq \frac{1}{2}c(T^2 - T_c^2)h^2 - \frac{eT}{12\pi}\left(h^2\right)^{3/2} + \frac{\lambda}{4}h^4$$

Note that at $T > T_c$, the mass term is positive: $SU(2)_L \times U(1)_Y$ is unbroken.

We therefore expect a phase transition at high temperatures in the early universe.

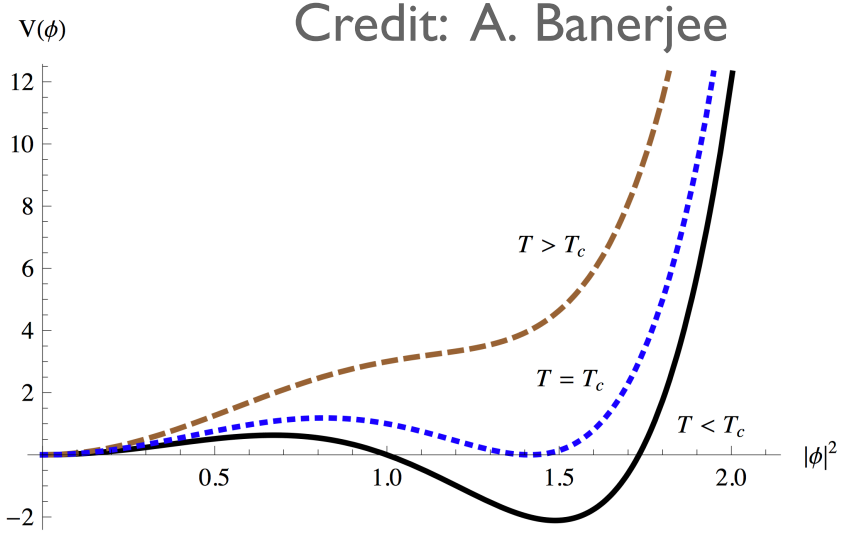
The Electroweak Phase Transition

The phase transition can be *second order* (driven by thermal fluctuations) or *first order* (starts with tunneling).



second order

Credit: A. Banerjee



first order

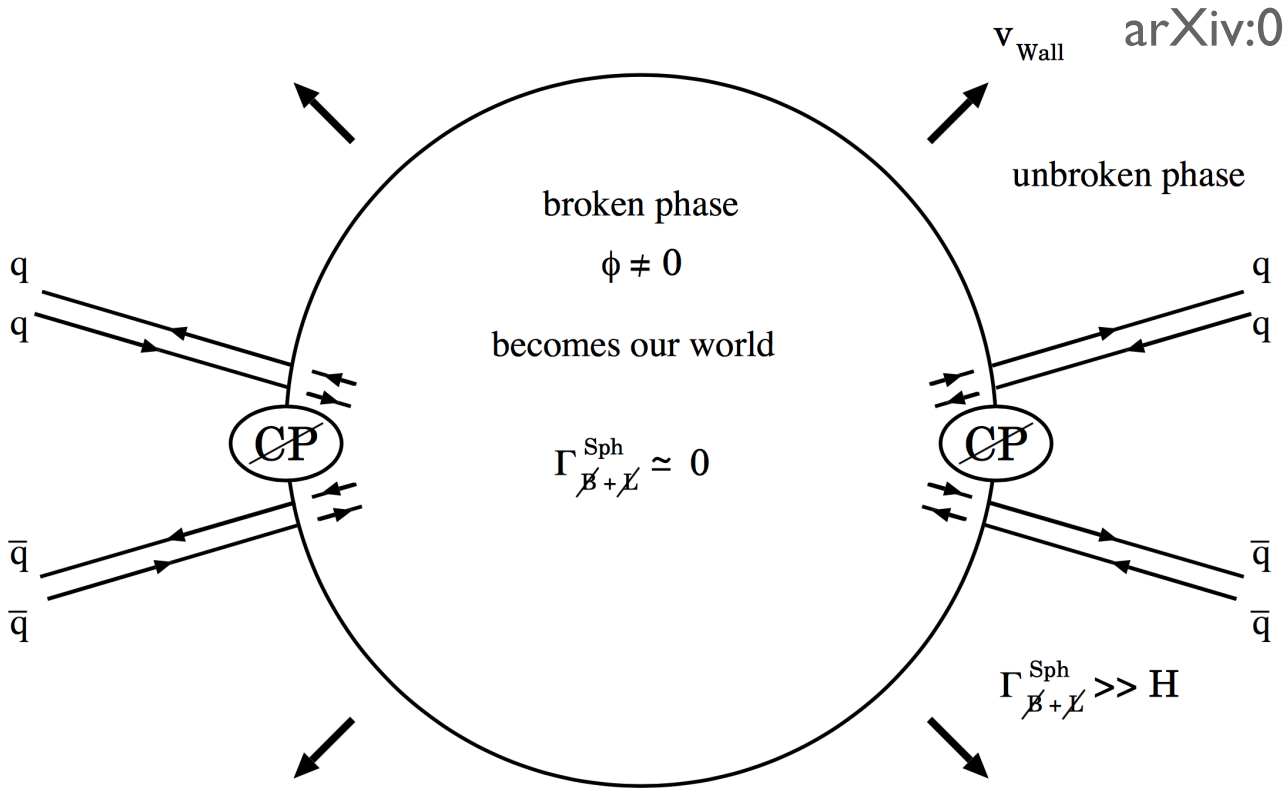
The barrier between phases is created by other particles coupled to the Higgs!

Baryogenesis at the Electroweak Phase Transition

For a strong first order transition tunneling nucleates bubbles that expand faster than the Hubble rate.

Processes at the bubble wall can lead to CP and Baryon number violation – potentially the source of the baryon asymmetry!

arXiv:0205279



Probing Baryogenesis via the Higgs Potential

The strength of the transition is determined by the shape of the potential, with the order parameter given by

$\sim h^3$ term

$$\frac{v(T_c)}{T_c} \cong \frac{(e/4\pi)}{\lambda}$$

Fixed to SM value

which is large enough only for $m_h \lesssim 48$ GeV.

To get a strong first order EWPT, we need additional contributions to the Higgs potential – these can be explored via colliders!

Probing the Higgs Potential

The Trillion Dollar Argument

Clearly we need to understand the Higgs potential beyond the mass term.

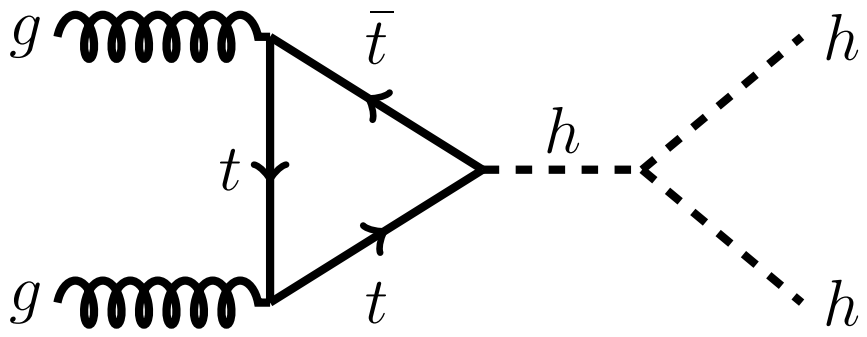
The only way to do this is to look for Higgs couplings to itself $\implies$ di-Higgs Production!

We could also look for $h \to hhh$, but this is much harder.

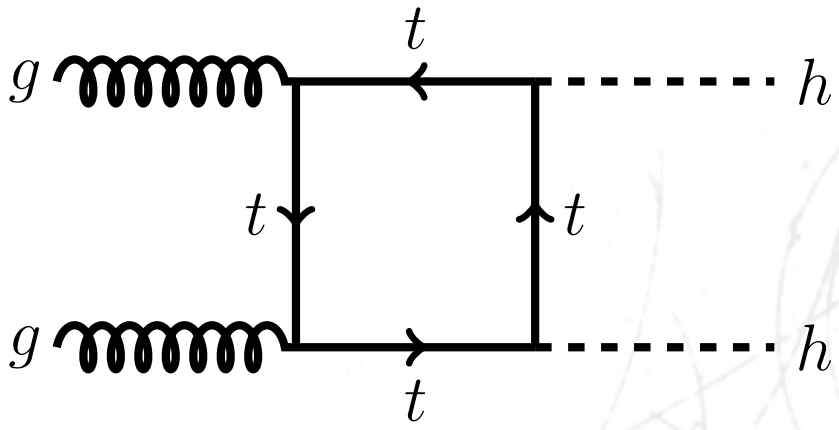
How do we look for di-Higgs production at colliders?

Di-Higgs Production

The λ_3 vertex contributes to di-Higgs production:



There's also an interference term:



Higgs Decay Modes

Higgs couples to particles proportional to their mass.

Final State	Branching Ratio
$b\bar{b}$	0.58
W^+W^-	0.215
gg	0.09
$\tau^+\tau^-$	0.06
ZZ	0.026
$\gamma\gamma$	0.00228

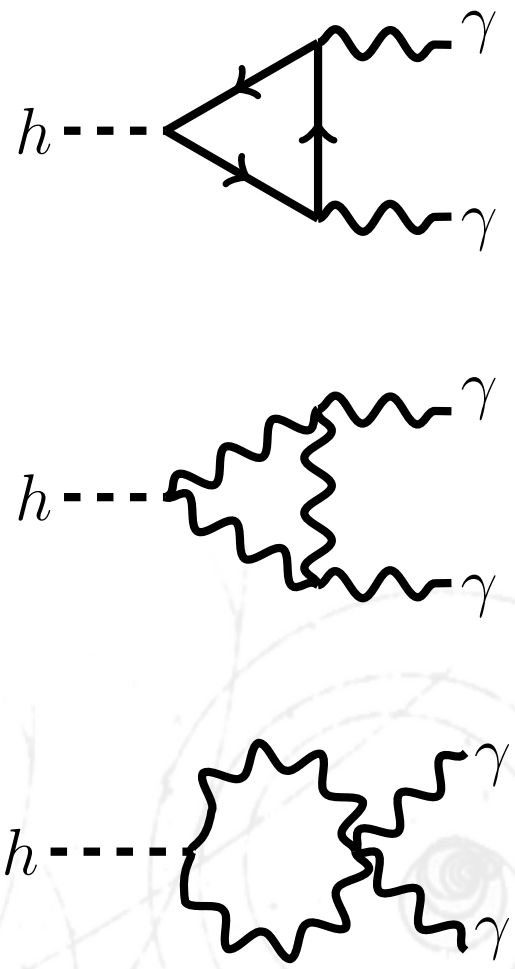
Table: Branching ratios of various decay modes for a 125 GeV Higgs boson.

Higgs Decay Modes

Loop induced couplings ($H \to gg$, $H \to \gamma\gamma$) are also important.

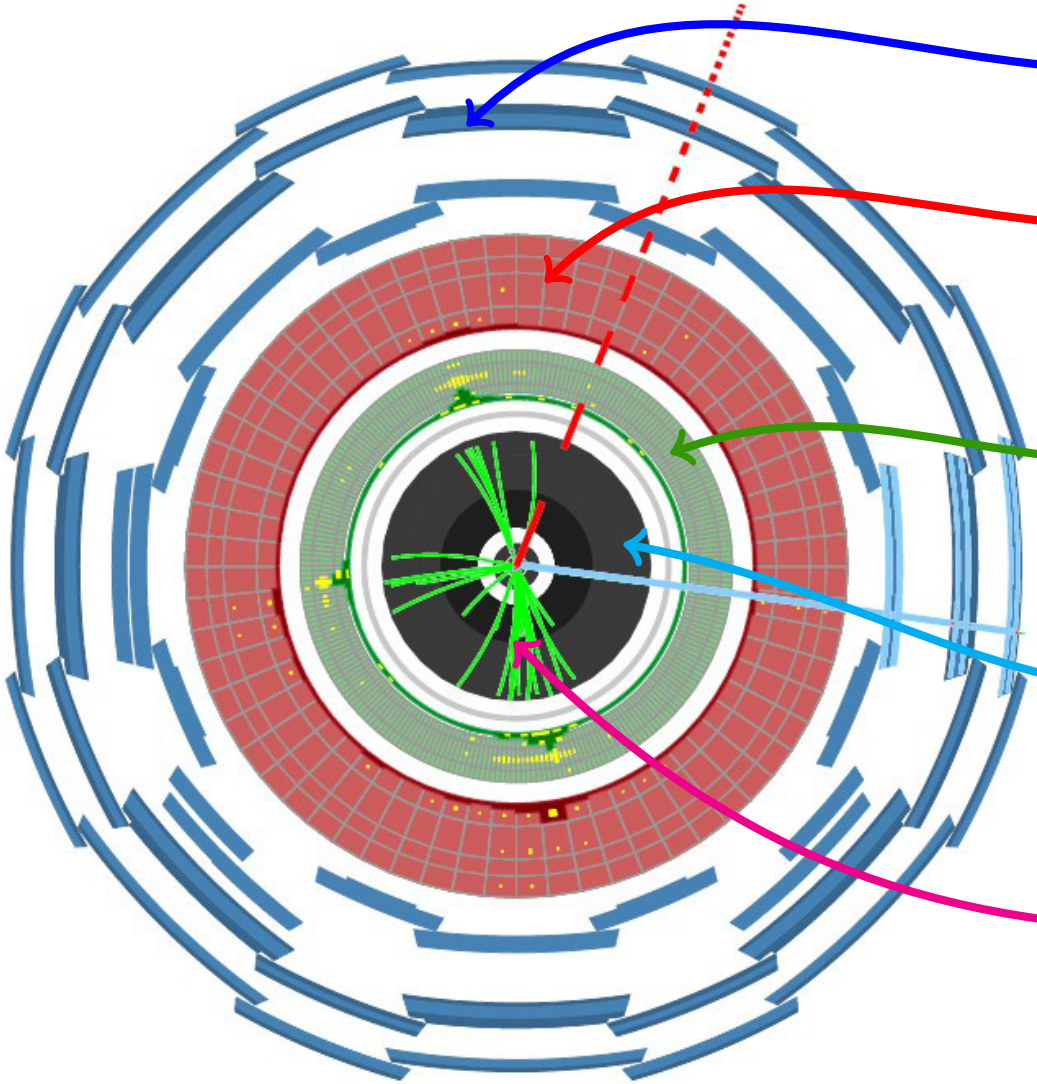
Diphoton channel has incredibly tiny branching ratio – 0.228% of decays!

But photon mass resolution is extremely good – the diphoton channel was essential to the original Higgs discovery.



Some LHC Phenomenology

(A Theorist's Understanding)



Muon Spectrometer

Hadronic Calorimeter
(Jets, etc.)

Electronic Calorimeter
(Photons & Electrons)

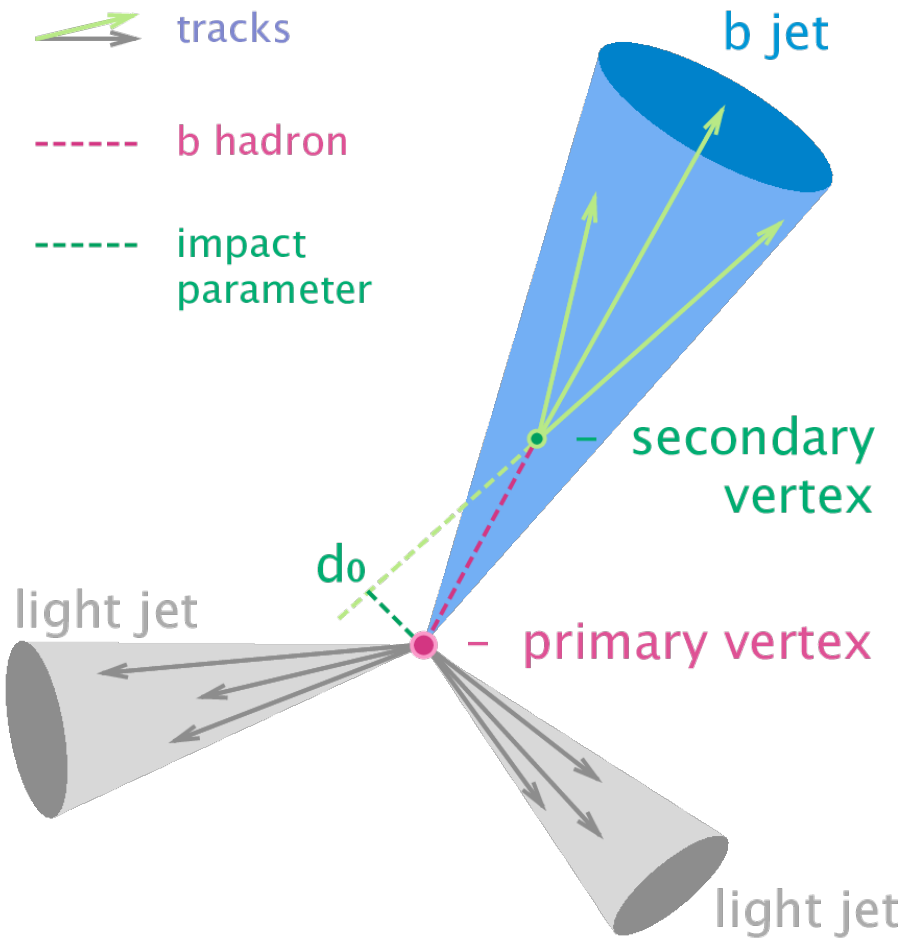
Tracker
(Four-Momenta via Magnets)

Pixel Detector
(Interaction Pts., Vertices)

atlasexperiment.org

B-Tagging

Making Sense of QCD Madness



Credit: Nazar Bartosik

b / $\bar{b}$ quarks form B-mesons, which have lifetimes $\sim 10^{-12}$ s

Travel a few millimeters before decaying!

Tagging efficiency roughly 75%, changes with p_T and η.

c-jets and light jets can sometimes be mis-tagged!

How do we search for two Higgs?

If One Wasn't Hard Enough

Two Higgs bosons have a variety of possible decay combinations:

- $hh \to b\bar{b}b\bar{b}$

  Dominant decay mode – 1/3 of events!

  Poor mass resolution – difficult to distinguish from ZZ

- $hh \to \gamma\gamma\gamma\gamma$

  Great mass resolution.

  $BR \sim 8 \times 10^{-6}$! Not even one will be produced at the LHC.

$hh \to b\bar{b}\gamma\gamma$

Best combination of mass resolution and large branching ratio!
The "golden channel".

Other Possibilities:
$hh \to (b\bar{b})(ZZ^*) \to (b\bar{b})(\ell^+\ell^-\ell^+\ell^-)$, $hh \to (b\bar{b})(\mu^+\mu^-)$
$hh \to (b\bar{b})(\tau^+\tau^-/W^+W^-)$, $hh \to (b\bar{b})(Z\gamma)$

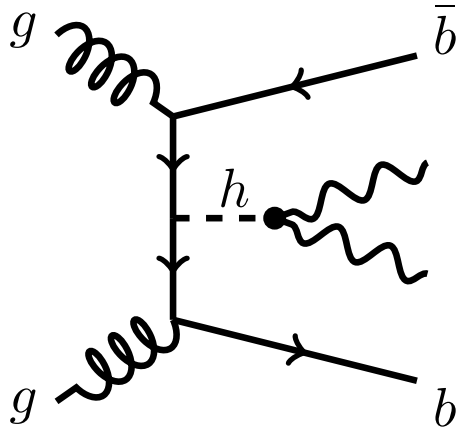
Search Strategy

Backgrounds

Now that we've chosen a decay channel, have to tabulate backgrounds.

What other processes lead to $b\bar{b}\gamma\gamma$?

- $b\bar{b}\gamma\gamma$
- $Z(\to b\bar{b}) + H(\to \gamma\gamma)$
- $b\bar{b} + H(\to \gamma\gamma)$
- $t\bar{t}(\to b\bar{b} + X) + H(\to \gamma\gamma)$



Note:

Ordinary single Higgs production is now a background!

Also other backgrounds from mistagging jets as b-jets, jets faking photons...

Search Strategy

Cut and Count..

Now that we've identified all the backgrounds, we have to play a game of statistics.

Come up with a set of cuts: which events do we want to count?

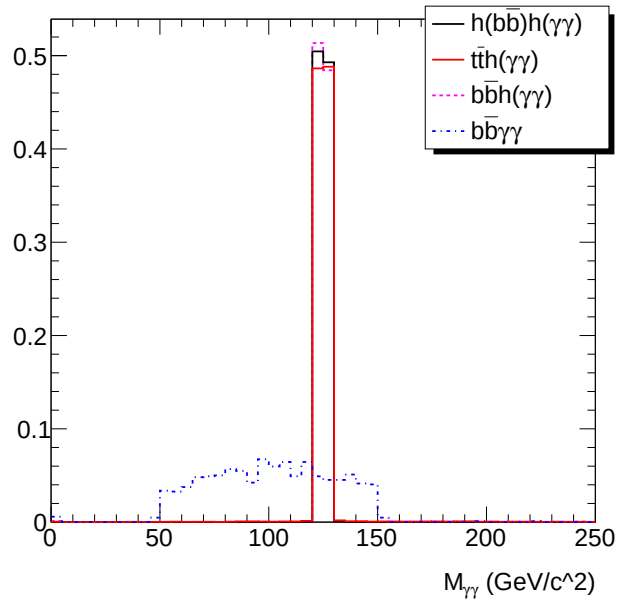
Once you've chosen cuts, count number of signal events you expect (S) and number of backgrounds (B).

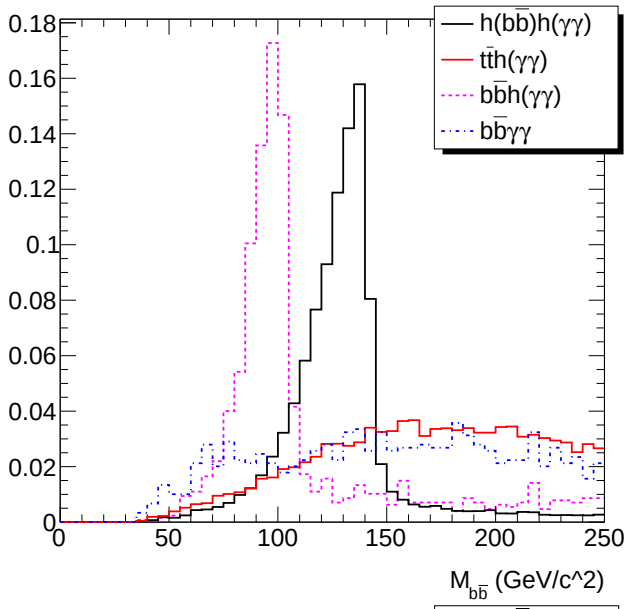
The "significance" is then $\sigma = S/\sqrt{S+B}$

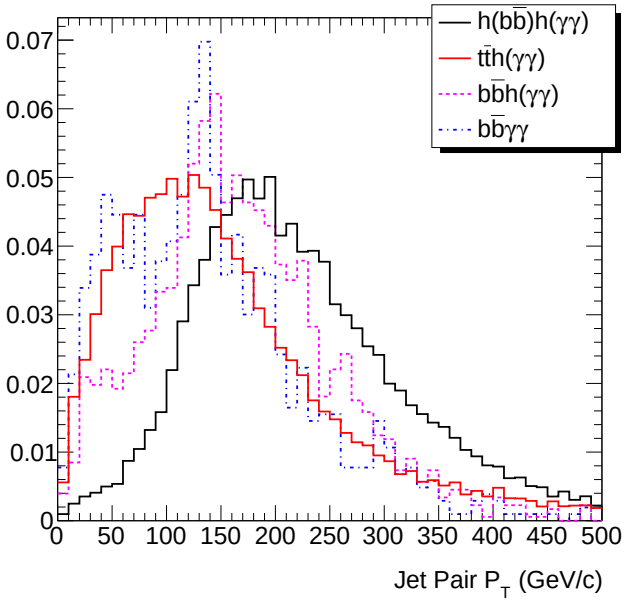
Goal: Come up with set of cuts that optimizes signal / background.

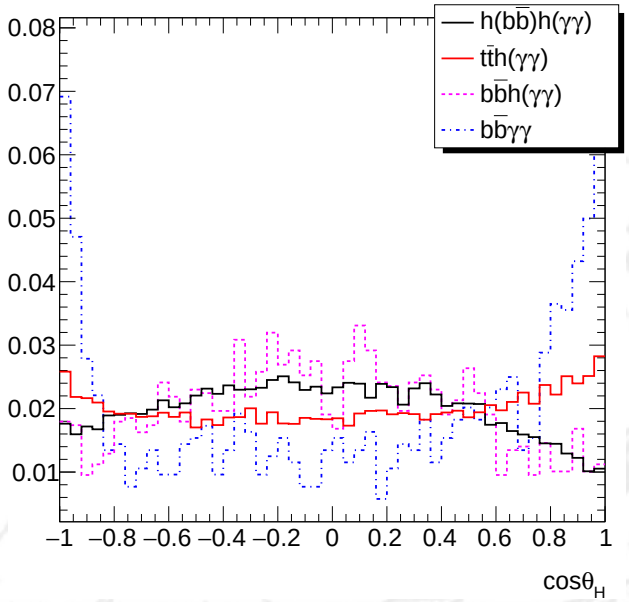
Simulations!

All the ROOT Plots

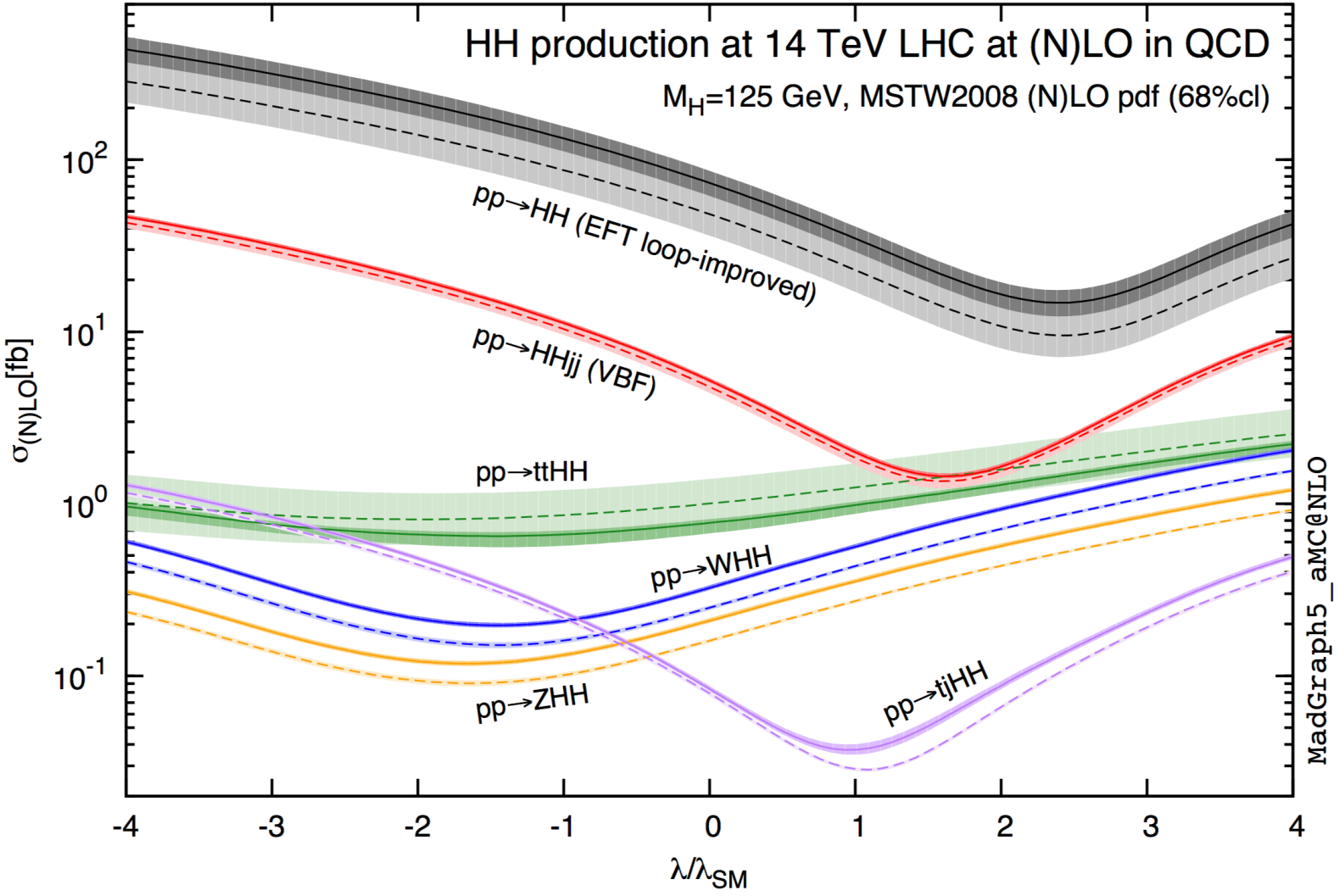








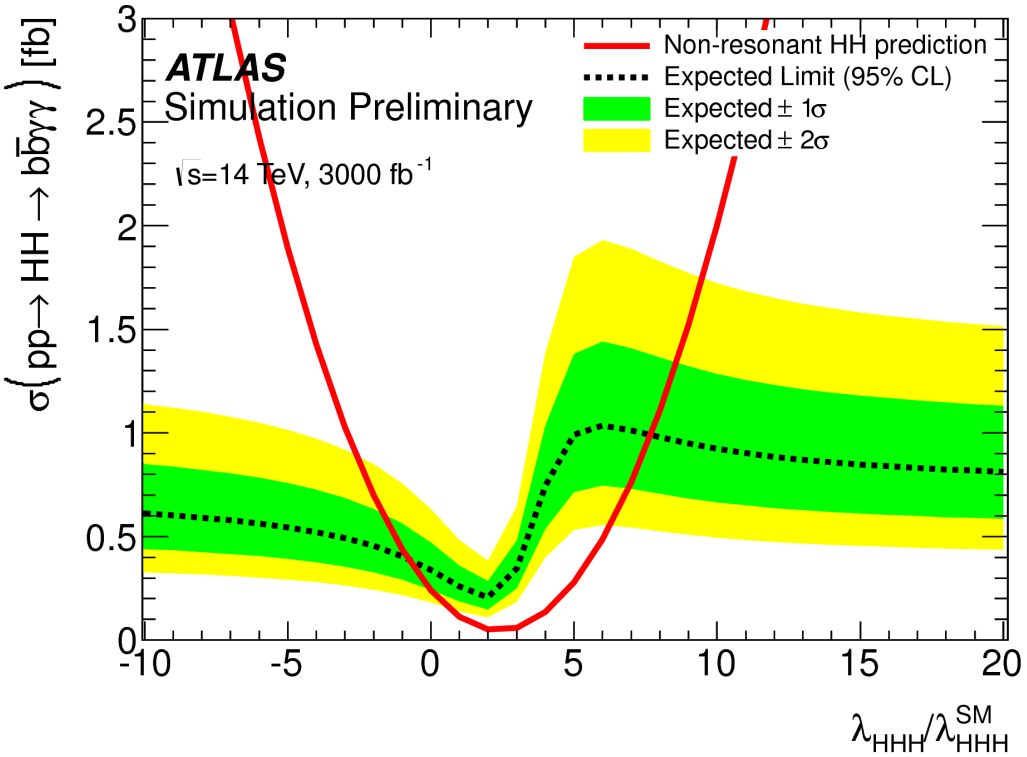
Extracting λ_3



arXiv:1401.7340

ATLAS Prospects at the HL-LHC

ATL-PHYS-PUB-2017-001



Expected constraint:

$$-0.8 < \lambda_3/\lambda_{3,SM} < 7.7$$

We need a higher energy collider! The FCC-hh is projected to improve this to $0.95 \lesssim \lambda_3/\lambda_{3,SM} \lesssim 1.05$.

Conclusions

- Measuring the Higgs trilinear coupling is an important probe of the consistency of the Standard Model.
- Understanding the Higgs sector gives us insight into the electroweak phase transition and cosmology at high temperatures.
- Precision measurements of λ_3 can be made by studying $gg \to hh \to b\bar{b}\gamma\gamma$ at high energy colliders.